The first five omission colours at a critical six-point head

Human Compression project: bounded L044 continuation

We retain the full standing hypotheses and explicit smaller-degree alternating induction of the accepted critical-head theorem. Thus $G \le A_n$ is transitive with connected string C-marking and all maximal omissions intransitive, $n \ge 12$, and the actual first three generators form an extendable six-point Table 2 head. Assume rank failure. That theorem gives $n = 2r \ge 18$ and, in the literal accepted orientation,

$$\begin{aligned} &D = \{1, \dots, 6\}, \quad E = \Omega \setminus D, \quad Z = E \cup \{6\}, \qquad G_{\ge 3} = \mathrm{Alt}(Z), \\ &\rho_0 = (1\ 2)(3\ 4), \quad \rho_1 = (2\ 3)(4\ 5), \\ &\rho_2 = (1\ 2)(5\ 6) \text{ or } (3\ 4)(5\ 6), \qquad \tau = \rho_3 = (6\ b)\delta, \quad b \in E. \end{aligned}$$

Here δ is odd and supported on $E \setminus \{b\}$. Put $\sigma = \rho_4$, $K = G_{\ge 5}$ and $B = G_{\ge 4}$. The group K fixes $D \cup \{b\}$ and centralizes δ; σ fixes D pointwise. No transitivity or natural alternating identification is assumed for B.

Lemma 1 (Colour four is never a unique crossing)**.** *The actual omission G_4 has at least three orbits. Consequently colour 4 has at least two ambient-eligible edges.*

Proof. Transitivity of $G_{\ge 3}$ gives $x = b^\sigma \ne b$; otherwise $\langle \tau, \sigma, K \rangle$ would preserve $\{6, b\}$. The complete actual pure-head overlap argument gives $\tau x = x$: conjugating the head first by τ and then by σ supplies pure $\mathrm{Alt}(D \cup \{b, x\})$ in $G_{\le 4}$. If $\tau x \ne x$, another overlap supplies pure A_4 in $G_{\le 4} \cap G_{\ge 3} = \langle \tau, \sigma \rangle$, contrary to its dihedral structure. In particular $\delta x = x$.

Let $P = G_{\le 3}$. It preserves $W = D \cup \{b\}$ and is transitive there; its exterior action is exactly $\langle \delta \rangle$. The group K fixes W pointwise. Thus W is one actual $G_4 = \langle P, K \rangle$ orbit, and the exterior orbits are those of $\langle \delta, K \rangle$. Since δ commutes with K and fixes x, the exterior orbit of x is exactly x^K and is pointwise fixed by δ. The odd involution δ is nonidentity and fixes all of W as well, so some moved point belongs to a third orbit.

Adding the edges of ρ_4 to the omission graph must connect all its orbits, by transitivity of G. A single eligible edge can connect only two of them. Hence there are at least two. This argument concerns full ambient omission orbits, not a local omission in a projected parabolic. □

Lemma 2 (The literal lower-colour routes)**.** *Colours 0 and 2 have unique eligible edges; colour 1 has two. Their exact omission orbits are*

	row 2 head	*row 3 head*
G_0	$\{1,2,3\},\ \Omega \setminus \{1,2,3\}$	$\{1\},\ \Omega \setminus \{1\}$
G_1	$\{1,2\},\ \{3,4\},\ E \cup \{5,6\}$	
G_2	$\{1,2,3,4,5\},\ Z$	

At colour 0, the row 2 head has right-shared labels exactly $1, 2$, while the row 3 head has none. The colour-2 cut is unshared. Colour 3 has a unique eligible edge if and only if B is transitive on E; if so, its two sides are the actual supported groups A_5 on D and B on E, and it is unshared.

Proof. The full tail $G_{\ge 3} = \mathrm{Alt}(Z)$ is transitive on Z and fixes the first five points. Add the displayed remaining literal head edges after each omission; they give exactly the listed components. For G_0, the eligible edge is $\{3, 4\}$ in row 2 and $\{1, 2\}$ in row 3. Both edges of ρ_1 join

different G_1-orbits. For G_2, only $\{5,6\}$ crosses. The displayed supports give all claims about shared labels: every label at least 3 fixes the first five points.

Finally $G_3 = \langle A_5, B\rangle$ is an actual product on disjoint supports D, E. Its orbits are D and all the B-orbits in E, including fixed points. If B is transitive, precisely the transposition $(6\ b)$ in τ joins the two orbits; all δ-edges remain in E. Conversely a unique eligible edge and transitivity of G force exactly two omission orbits, so B is transitive. All prefix labels are confined to D and all suffix labels to E, proving that this cut is unshared. □

Scope and source route. These are exact early-colour adapters for the exceptional-head continuation of source Propositions 5.28–5.32, physical pages 36–45. The literal Table 2 data and the accepted critical-tail and pure-overlap proofs remain paid. The new third-orbit argument excludes a unique colour-4 crossing even when B is intransitive. It does not make B natural alternating, eliminate a shared cut of higher colour, or resolve the remaining full-source branches. No new finite computation, formalization, original or question credit is claimed.